

SIH26249 — 06 Demo Script & Judge Q&A

SIH 2026 — SIH26249

Air Power: Predictive Maintenance & Fleet Availability

Ministry of Defence — Defence Services Staff College

06 — Demo Script & Judge Q&A Armoury

HONESTY RULES — READ BEFORE THE DEMO (shown to the team, enforced on screen)

1. **ALL data is synthetic.** "SYNTHETIC DATA" banner on **every screen**, no exceptions.
2. Every API response carries `"data_class": "synthetic"` — open `/docs` if a judge asks.
3. **Never imply real IAF fleet data.** Synthetic tail numbers only (e.g., `SYN-01`); no real squadrons, no roundels.
4. **Never claim "worldwide first"** — the classified space is unknowable. Say "not advertised in their public product material."
5. **We are the decision layer, never a competing sensor/HUMS box.** HAL's HUMS does per-platform diagnostics; we sit ABOVE it.
6. **Thin-evidence discipline:** GE numbers only with "per GE"; Air Chief quotes only with "per press reports"; Safran scale figures are ~2019 press-release era — never current; **never cite the AMCA-forum "75% MC rate" as an IAF target**; no primary source confirms an AI-RUL HAL ground station — do not claim it.
7. **Measured vs ESTIMATE:** every number on screen is either computed live from the synthetic pipeline or labeled ESTIMATE. Nothing in the deck claims a number not in `evidence/`.

On-screen honesty badges (visible during the whole demo)

- Top banner, every screen: **"SYNTHETIC DATA — demonstration only, not real IAF fleet data"**
- Forecast chart footer: "Forecast from synthetic RUL + spares + crew + demand model · seed #____ · recomputed live"
- Eval printout slide: "Measured on held-out synthetic tails · does not transfer to real aircraft"

5-MINUTE DEMO SCRIPT (minute by minute)

0:00 — Problem + USP. Read the PS line: maintenance data "not adequately integrated" → delayed prediction, avoidable downtime. "Every tool predicts component health. Nobody tells the commander how many aircraft fly tomorrow. We built that layer — the fleet-availability decision layer."

0:45 — Fleet board. 40 synthetic tails, MC/PMC/NMC status colors, SYNTHETIC banner visible. "This is the commander's morning screen — one glance, whole fleet."

1:30 — Tail drill-down + explainability. Click an NMC tail: components, RUL 38 h [31–45 h], top contributing features, plain-English reason. "Not a black box — every prediction shows its work."

2:15 — MC forecast curve + unfilled sorties. 7-day forecast; headline: "Thursday: 31 of 40 mission-capable, 3 high-priority sorties unfilled — that gap is the mission impact."

3:00 — What-if levers, LIVE. "Expedite these two parts, reassign one crew shift." Click apply → before/after MC delta recomputed in front of them (< 200 ms). "The action today that changes tomorrow's number."

3:45 — Cannibalization advisor + debt ledger. NMC recipient, proposed donor (lower priority, already NMC — guardrails shown passing), net MC effect, debt entry created with repayment queue. "Authorized, guardrailed, repaid — never a shadow practice."

4:15 — Audit log + evidence. Hash-chained log of everything just done. Then the `eval.py` printout: "RUL model RMSE __ vs naive baseline __, measured on held-out synthetic tails. This is our evidence discipline — measured, not claimed."

4:45 — Integration roadmap + closing. "The roadmap is integration into IAF maintenance IT — Maintenix/Ramco records, HUMS feeds. We sit above all of them." **Closing line: "We don't predict engine failure — we tell the commander how many aircraft fly tomorrow."**

THE 10 TOUGHEST JUDGE QUESTIONS (one-line rebuttals)

1. **"HAL already runs HUMS with AI RUL — what's new?"** → HAL HUMS is per-platform component diagnostics; we sit *above* it, fusing health with spares, crew, records, and sortie demand into a fleet availability forecast. (No primary source confirms an AI-RUL HAL ground station — do not concede it as fact.)
2. **"Why not LSTM on C-MAPSS like everyone?"** → RUL is the commodity; the winning layer is the decision layer. GBM-on-windowed-features matches the published accuracy band, trains in seconds on CPU, and retrains live.
3. **"Synthetic data — does this work on real aircraft?"** → Honest answer: we don't claim it does. This is a validated decision-loop prototype; the roadmap is integration into IAF maintenance IT. The generator is seeded, deterministic, and documented — here is the SHA-256 proof.
4. **"GE/Safran already do predictive maintenance."** → Engine-centric, commercial, OEM-locked *per their own product pages*; no fleet-level mission-capable forecast, no what-if levers, no cross-OEM fusion.
5. **"DRDO has a digital-twin RFP for this."** → An asset-level engine-twin RFP with no delivered product evidenced; our layer *consumes* twin outputs as inputs — complementary, not competing.
6. **"Cannibalization is bad practice."** → GAO treats it as a symptom of logistics shortfalls; our advisor wraps it in guardrails — lower-priority NMC donors only, CANN-rate limits, a debt ledger with a repayment queue. Authorized, tracked, repaid.
7. **"Ramco 6.0 already has AI predictive maintenance."** → Descriptive and assistive AI on records; no availability decision layer; their AI metrics are vendor claims with no independent evaluation.
8. **"What stops another team copying this?"** → The fusion logic, what-if engine, and audit trail are a systems build, not a model — and our evidence discipline (measured RMSE vs baseline, live before/after demo) is the moat.

9. **"How is this different from aerotwin-ai (SIH26054)?"** → Different PS entirely — DRDO MALE-UAV piston engines, an engine-twin dashboard; no fleet-availability logic evidenced. Related work, not a competitor.
 10. **"Scale: one base vs the whole IAF?"** → Demo is one base on one laptop by design; the architecture shows the path — SQLite→Postgres swap, multi-base federation as documented future work.
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Backup answers (if they dig deeper)

- **"Where do thresholds come from?"** → One config file, `config/thresholds.yaml` — open it live. All doctrine-mapped, all adjustable.
- **"Why greedy, not MILP?"** → 36-hour honesty: a full MILP fleet optimizer isn't feasible in the build window; the greedy net-MC heuristic with guardrails is the demo, OR-Tools MILP is the documented stretch.
- **"What about data security / real IAF data?"** → We never touch it. The product is designed to consume feeds inside IAF maintenance IT; our demo is 100% synthetic by architecture, not by promise.